

Assignment-02

Bootstrap 5

1.Solution for Question 1:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Bootstrap</title>
  <link rel="stylesheet" href="../../bootstrap-5.3.8-dist/css/bootstrap.min.css">
  <style>
    .custom-paragraph {
      text-align: justify;
      font-size: 1.25rem;
      color: #2c3e50;
      line-height: 1.8;
    }
  </style>
</head>
<body class="bg-light">
  <div class="mb-4">
    <h1 class="display-4 text-white bg-primary text-center p-4 m-0">Bootstrap: A CSS
framework</h1>
  </div>
  <div class="container">
    <section class="mb-4">
      <p class="custom-paragraph p-4 shadow-sm bg-white rounded-3">
        Bootstrap is a free and open-source CSS framework directed at responsive, mobile-first
front-end web development. It contains HTML, CSS and JavaScript-based design templates for
typography, forms, buttons, navigation, and other interface components. It is one of the most popular
front-end frameworks in the world, enabling developers to build visually appealing and highly
functional websites quickly and efficiently.
      </p>
    </section>
    <section class="mb-5">
      <div class="text-end">
        <a href="#" class="btn btn-primary btn-lg px-4">Read More</a>
      </div>
    </section>
    <section class="mb-5">
      <h3 class="text-center mb-3">Bootstrap Version History</h3>
      <table class="table table-bordered table-striped table-hover shadow-sm bg-white w-50 mx-
auto">
        <thead class="table-dark">
          <tr>
            <th>Bootstrap Version</th>
            <th>Release Date</th>
            <th>Major Features</th>
          </tr>
        </thead>
        <tbody>
```

```

<tr>
  <td>Bootstrap 1</td>
  <td>August 19, 2011</td>
  <td>
    <ul>
      <li>maintained by Otto</li>
      <li>Thornton, a small group of core developers</li>
      <li>a large community of contributors</li>
    </ul>
  </td>
</tr>
<tr>
  <td>Bootstrap 2</td>
  <td>January 31, 2012</td>
  <td>
    <ul>
      <li>built-in support for Glyphicons</li>
      <li>dynamically, taking into account</li>
      <li>Otto and Thornton left Twitter</li>
    </ul>
  </td>
</tr>
<tr>
  <td>Bootstrap 3</td>
  <td>August 19, 2013</td>
  <td>
    <ul>
      <li>Mobile-first design philosophy</li>
      <li>Flat design adoption</li>
      <li>New grid system</li>
    </ul>
  </td>
</tr>
<tr>
  <td>Bootstrap 4</td>
  <td>January 18, 2018</td>
  <td>
    <ul>
      <li>Switched to Sass from Less</li>
      <li>Flexbox support enabled</li>
      <li>Dropped IE8, IE9, and iOS 6 support</li>
    </ul>
  </td>
</tr>
<tr>
  <td>Bootstrap 5</td>
  <td>May 5, 2021</td>
  <td>
    <ul>
      <li>jQuery dependency removed</li>
      <li>Custom CSS properties added</li>
      <li>Offcanvas component introduced</li>
    </ul>
  </td>
</tr>

```

```

        </tbody>
    </table>
</section>

<section class="mb-5">
    <h3 class="text-center mb-4">Latest Blogs</h3>
    <div class="row g-4">
        <div class="col-12 col-md-6 col-lg-4">
            <div class="card h-100 shadow-sm">
                
                <div class="card-body d-flex flex-column">
                    <h5 class="card-title">Understanding Grid Systems</h5>
                    <p class="card-text">Learn how the 12-column grid system works in Bootstrap to
create fully responsive layouts.</p>
                    <a href="#" class="btn btn-primary mt-auto align-self-start">Read More</a>
                </div>
            </div>
        </div>

        <div class="col-12 col-md-6 col-lg-4">
            <div class="card h-100 shadow-sm">
                
                <div class="card-body d-flex flex-column">
                    <h5 class="card-title">Bootstrap 5 Forms</h5>
                    <p class="card-text">A complete guide to designing beautiful and accessible forms
using Bootstrap 5 classes.</p>
                    <a href="#" class="btn btn-primary mt-auto align-self-start">Read More</a>
                </div>
            </div>
        </div>

        <div class="col-12 col-md-6 col-lg-4">
            <div class="card h-100 shadow-sm">
                
                <div class="card-body d-flex flex-column">
                    <h5 class="card-title">Customizing with Sass</h5>
                    <p class="card-text">How to override default Bootstrap variables and compile your
own custom stylesheets using Sass.</p>
                    <a href="#" class="btn btn-primary mt-auto align-self-start">Read More</a>
                </div>
            </div>
        </div>
    </div>
</section>

<section class="mb-5">
    <div class="row justify-content-center">
        <div class="col-12 col-md-8 col-lg-6">
            <div class="card shadow">
                <div class="card-header bg-dark text-white text-center py-3">
                    <h4 class="mb-0">Create an Account</h4>
                </div>
            </div>
        </div>
    </div>
</section>

```

```

<div class="card-body p-4">
  <form>
    <div class="mb-3">
      <label for="inputName" class="form-label">Full Name</label>
      <input type="text" class="form-control" id="inputName" placeholder="John
Doe" required>
    </div>

    <div class="mb-3">
      <label for="selectGender" class="form-label">Gender</label>
      <select class="form-select" id="selectGender" required>
        <option value="" selected disabled>Choose...</option>
        <option value="male">Male</option>
        <option value="female">Female</option>
        <option value="other">Other</option>
      </select>
    </div>

    <div class="mb-3">
      <label for="inputDOB" class="form-label">Date of Birth</label>
      <input type="date" class="form-control" id="inputDOB" required>
    </div>

    <div class="mb-3">
      <label for="inputAddress" class="form-label">Address</label>
      <textarea class="form-control" id="inputAddress" rows="3" placeholder="123
Main St..." required></textarea>
    </div>

    <div class="mb-3">
      <label for="inputMobile" class="form-label">Mobile Number</label>
      <input type="tel" class="form-control" id="inputMobile" placeholder="+91
XXXXXX XXXXXX" required>
    </div>

    <div class="mb-3">
      <label for="inputEmail" class="form-label">Email Address</label>
      <input type="email" class="form-control" id="inputEmail"
placeholder="name@example.com" required>
    </div>

    <div class="mb-4">
      <label for="inputPassword" class="form-label">Password</label>
      <input type="password" class="form-control" id="inputPassword" required>
    </div>

    <button type="submit" class="btn btn-primary w-100 py-2">Sign Up</button>
  </form>
</div>
</div>
</div>
</div>
</section>
</div>
<script src="../../bootstrap-5.3.8-dist/js/bootstrap.bundle.min.js"></script>

```

```
</body>  
</html>
```

Output:

Desktop view:

Bootstrap: A CSS framework

Bootstrap is a free and open-source CSS framework directed at responsive, mobile-first front-end web development. It contains HTML, CSS and JavaScript-based design templates for typography, forms, buttons, navigation, and other interface components. It is one of the most popular front-end frameworks in the world, enabling developers to build visually appealing and highly functional websites quickly and efficiently.

[Read More](#)

Bootstrap Version History

Bootstrap Version	Release Date	Major Features
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Bootstrap 2	January 31, 2012	- built-in support for Glyphicons - dynamically, taking into account - Otto and Thornton left Twitter
Bootstrap 3	August 19, 2013	- Mobile-first design philosophy - Flat design adoption - New grid system
Bootstrap 4	January 18, 2018	- Switched to Sass from Less - Flexbox support enabled - Dropped IE8, IE9, and iOS 6 support
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Latest Blogs



Understanding Grid Systems

Learn how the 12-column grid system works in Bootstrap to create fully responsive layouts.

[Read More](#)

Bootstrap 5 Forms

A complete guide to designing beautiful and accessible forms using Bootstrap 5 classes.

[Read More](#)

Customizing with Sass

How to override default Bootstrap variables and compile your own custom stylesheets using Sass.

[Read More](#)

Create an Account

Full Name

John Doe

Gender

Choose... ▾

Date of Birth

dd-mm-yyyy 

Address

123 Main St... 

Mobile Number

+91 XXXXX XXXXX

Email Address

name@example.com

Password

Sign Up

Mobile view:

Bootstrap: A CSS framework

Bootstrap is a free and open-source CSS framework directed at responsive, mobile-first front-end web development. It contains HTML, CSS and JavaScript-based design templates for typography, forms, buttons, navigation, and other interface components. It is one of the most popular front-end frameworks in the world, enabling developers to build visually appealing and highly functional websites quickly and efficiently.

[Read More](#)

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How to override default Bootstrap variables and compile your own custom stylesheets using Sass.

[Read More](#)

Create an Account

Full Name

John Doe

Gender

Choose...

Date of Birth

dd/mm/yyyy

Address

123 Main St...

Mobile Number

+91 XXXXX XXXXX

Email Address

name@example.com

Password

[Sign Up](#)

2.Solution for Question 2:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Suman Portfolio</title>
  <link rel="stylesheet" href="../bootstrap-5.3.8-dist/css/bootstrap.min.css">
  <link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/bootstrap-icons@1.13.1/font/bootstrap-
icons.min.css">
</head>
<body class="bg-black">
  <div class="container">
    <div class="row">
      <!-- Navbar -->
      <nav class="navbar navbar-expand-lg ">
        <div class="container-fluid">
          <a class="navbar-brand text-light" href="#">Suman Nayak</a>
          <button class="navbar-toggler" type="button" data-bs-toggle="collapse" data-bs-
target="#navbarNav"
          aria-controls="navbarNav" aria-expanded="false" aria-label="Toggle navigation">
            <span class="navbar-toggler-icon"></span>
          </button>
          <div class="collapse navbar-collapse" id="navbarNav">
            <ul class="navbar-nav ms-auto ">
              <li class="nav-item">
                <a class="nav-link text-light" aria-current="page" href="#">Home</a>
              </li>
              <li class="nav-item">
                <a class="nav-link text-light" href="#about">About</a>
              </li>
              <li class="nav-item">
                <a class="nav-link text-light" href="#academics">Academics</a>
              </li>
              <li class="nav-item">
                <a class="nav-link text-light" href="#skills">Skill</a>
              </li>
              <li class="nav-item">
                <a class="nav-link text-light " href="#projects">Projects</a>
              </li>
            </ul>
          </div>
        </div>
      </nav>
      <!-- Hero Section -->
      <header class="mt-5">
        <div class="container">
          <div class="row g-4 align-items-center">
            <div class="col-4 col-lg-6 text-center">
              
            </div>
            <div class="col-md-6 text-light">
```



```

<span class="badge bg-info text-dark p-3">Hi, I am</span>
<h1 class="display-4 mt-2">Suman Nayak</h1>
<h6>Master of Computer Application</h6>
<p class="mt-3 mb-2">
  <i class="bi bi-envelope"></i>suman.work.in@gmail.com
</p>
<p class="mb-2">
  <i class="bi bi-phone"></i>+91 7077262619
</p>
<p class="mb-2">
  <i class="bi bi-linkedin"></i>linkedin.com/in/suman-nayak-b7b582391/
</p>
<p>
  <i class="bi bi-github"></i>github.com/suman-nayak
</p>
</div>
</div>
</div>
</header>
<!-- About Page -->
<section id="about" class="container mt-5 pt-4">
  <div class="row text-light align-items-center">
    <div class="col-md-6">
      <h1 class="text-center mb-3">About Me</h1>
      <p class="custom-justify">
        Hello! I am Suman Nayak, an enthusiastic and motivated technology learner with a
        strong interest in Cloud Computing, DevOps, Python, and Web Development.
        I enjoy learning how modern applications are developed, deployed, and managed using
        cloud technologies and automation tools. I have been building my skills in AWS, Linux, Python,
        HTML, CSS, Git, CI/CD, and Infrastructure as Code, while working on practical projects to
        strengthen my technical knowledge.
        I am a quick learner, problem solver, and continuous learner who enjoys taking on new
        challenges and improving my skills through hands-on practice. My goal is to begin my career in a
        Cloud/DevOps or Software Development role, contribute to real-world projects, and grow into a
        skilled technology professional.
        Outside of coding, I believe in continuous improvement, teamwork, and staying
        curious about emerging technologies.
      </p>
      <button type="button" class="btn btn-info">
        <a href="#" target="_blank">Download CV</a>
      </button>
    </div>
    <div class="col-md-6 text-center d-none d-md-block">
      
    </div>
  </div>
</section>
<!-- Academic Section -->
<section id="academics" class="container mt-5 pt-4">
  <div class="row text-light g-3">
    <h1 class="text-center mb-3">My Academic Records</h1>
    <div class="col-md-4">
      <div class="card h-100">
        <div class="card-body">
          <h5 class="card-title">MCA</h5>

```

```

        <p class="card-text mb-1">
            <i class="bi bi-mortarboard"></i>Silicon University
        </p>
        <p class="card-text mb-1">
            <i class="bi bi-pencil"></i> 6.91 (CGPA)
        </p>
        <p class="card-text mb-0">
            <i class="bi bi-calendar-check"></i>2025 - 2027
        </p>
    </div>
</div>
</div>
<div class="col-md-4">
    <div class="card h-100">
        <div class="card-body">
            <h5 class="card-title">B.Sc(Physics)</h5>
            <p class="card-text mb-1">
                <i class="bi bi-mortarboard"></i>F.M University
            </p>
            <p class="card-text mb-1">
                <i class="bi bi-pencil"></i>7.71 (CGPA)
            </p>
            <p class="card-text mb-0">
                <i class="bi bi-calendar-check"></i>2021 - 2024
            </p>
        </div>
    </div>
</div>
<div class="col-md-4">
    <div class="card h-100">
        <div class="card-body">
            <h5 class="card-title">Higher Secondary</h5>
            <p class="card-text mb-1">
                <i class="bi bi-mortarboard"></i>Bhadrak Autonomous College
            </p>
            <p class="card-text mb-1">
                <i class="bi bi-pencil"></i>77.05%
            </p>
            <p class="card-text mb-0">
                <i class="bi bi-calendar-check"></i>2019 - 2021
            </p>
        </div>
    </div>
</div>
<div class="col-md-4">
    <div class="card h-100">
        <div class="card-body">
            <h5 class="card-title">Secondary</h5>
            <p class="card-text mb-1">
                <i class="bi bi-mortarboard"></i>A.C.B.N Bhadrak
            </p>
            <p class="card-text mb-1">
                <i class="bi bi-pencil"></i>79.09%
            </p>
            <p class="card-text mb-0">

```

```

        <i class="bi bi-calendar-check"></i>2019
    </p>
</div>
</div>
</div>
</div>
</div>
</section>

<!-- Technical Skills -->
<section id="skills" class="container mt-5 pt-4">
    <div class="row text-light g-3">
        <h1 class="text-center mb-3">My Technical Skills </h1>
        <div class="col-md-6">
            <p class="mb-1">Python</p>
            <div class="progress mb-2" role="progressbar" aria-valuenow="60" aria-valuemin="0"
                aria-valuemax="100">
                <div class="progress-bar bg-primary" style="width: 80%">80%</div>
            </div>

            <p class="mb-1">C</p>
            <div class="progress mb-2" role="progressbar" aria-valuenow="75" aria-valuemin="0"
                aria-valuemax="100">
                <div class="progress-bar bg-primary" style="width: 75%">75%
                </div>
            </div>

            <p class="mb-1">Java</p>
            <div class="progress mb-2" role="progressbar" aria-valuenow="70" aria-valuemin="0"
                aria-valuemax="100">
                <div class="progress-bar bg-primary" style="width: 78%">78%</div>
            </div>

            <p class="mb-1">PHP</p>
            <div class="progress mb-2" role="progressbar" aria-valuenow="75" aria-valuemin="0"
                aria-valuemax="100">
                <div class="progress-bar bg-primary" style="width: 75%">75%</div>
            </div>
        </div>

        <div class="col-md-6">
            <p class="mb-1">HTML</p>
            <div class="progress mb-2" role="progressbar" aria-valuenow="90" aria-valuemin="0"
                aria-valuemax="100">
                <div class="progress-bar bg-primary" style="width: 90%">90%</div>
            </div>

            <p class="mb-1">CSS</p>
            <div class="progress mb-2" role="progressbar" aria-valuenow="65" aria-valuemin="0"
                aria-valuemax="100">
                <div class="progress-bar bg-primary" style="width: 65%">65%</div>
            </div>

            <p class="mb-1">JavaScript</p>
            <div class="progress mb-2" role="progressbar" aria-valuenow="55" aria-valuemin="0"
                aria-valuemax="100">

```

```

        <div class="progress-bar bg-primary" style="width: 55%">55%</div>
    </div>

    <p class="mb-1">Bootstrap</p>
    <div class="progress mb-2" role="progressbar" aria-valuenow="85" aria-valuemin="0"
        aria-valuemax="100">
        <div class="progress-bar bg-primary" style="width: 85%">85%</div>
    </div>
</div>
</div>
</section>
<!-- Projects -->
<section id="projects" class="container mt-5 pt-4 text-light">
    <div class="row g-4">
        <div class="col-md-6">
            <div class="card bg-dark text-light h-100">
                <div class="card-body">
                    <h5 class="card-title">Real-Time Weather Dashboard</h5>
                    <p class="card-text mb-1">
                        <i class="bi bi-tools"></i>
                        Tools: Java 21, Java Swings, Maven, Java HttpClient, OpenWeatherMap
                        API, PostgreSQL
                    </p>
                    <p class="card-text">
                        A dynamic desktop application that integrates with RESTful APIs to fetch and
                        display live meteorological data. Engineered using JavaSwing for a responsive, modern UI and Java's
                        native HttpClient for efficient network requests. Implemented JSON parsing to extract complex data
                        structures, mapping live metrics (temperature, humidity, wind speed) to visual UI components and
                        condition-based dynamic backgrounds.
                    </p>
                </div>
            </div>
        </div>
        <div class="col-md-6">
            <div class="card bg-dark text-light h-100">
                <div class="card-body">
                    <h5 class="card-title">Secure Two-Tier Application Infrastructure on AWS</h5>
                    <p class="card-text mb-1">
                        <i class="bi bi-tools"></i>
                        Tools: AWS, Amazon EC2, Amazon VPC, AWS Client VPN Endpoint, ACM,
                        OpenVPN, PostgreSQL
                    </p>
                    <p class="card-text">
                        Your organization wants to host a web application on AWS. The web server
                        should be accessible from the internet, while the database server must remain private and inaccessible
                        from the public internet. Administrators should be able to securely connect to the private database
                        from their local machines using a VPN.
                    </p>
                </div>
            </div>
        </div>
    </div>
</section>

```

```

<div class="card-body">
  <h5 class="card-title">
    E-Commerce Website</h5>
  <p class="card-text mb-1">
    <i class="bi bi-tools"></i>
    Tools:Python, Django, SQLite, HTML & CSS, Bootstrap,JavaScript
  </p>

  <p class="card-text">
    This project is a comprehensive, full-stack e-commerce web application designed
    to simulate a real-world online shopping experience. It features a fully responsive, custom-themed
    user interface and handles the complete customer journey. Core functionalities include secure user
    authentication, dynamic product browsing, real-time cart management, and order history tracking.
    Additionally, it integrates a mock UPI payment gateway to seamlessly demonstrate the checkout and
    order-completion lifecycle.
  </p>
</div>
</div>
<div class="col-md-6">
  <div class="card bg-dark text-light h-100">
    <div class="card-body">
      <h5 class="card-title">Data Redundancy Removal System Design</h5>
      <p class="card-text mb-1">
        <i class="bi bi-tools"></i>
        Tools:Python, Flask, SQLite, Pandas, Git & GitHub
      </p>
      <p class="card-text">
        It acts as a smart gatekeeper for your database. When data arrives, the system
        extracts specific identifying fields and generates a unique cryptographic hash. By enforcing a strict
        database constraint on this hash, the pipeline accurately classifies incoming data as either unique, a
        "false positive" (similar data from a different source), or a true redundant duplicate. This ensures your
        database remains 100% accurate, highly efficient, and completely free of duplicate bloat.
      </p>
    </div>
  </div>
</div>
</div>
</section>
<!-- Footer -->
<footer class="container text-center text-light py-5">
  <p class="mb-0">
    &copy; 2026 Suman Nayak
  </p>
</footer>
<script src="../../bootstrap-5.3.8-dist/js/bootstrap.bundle.min.js"></script>
</body>
</html>

```

Output:

Desktop view:

Suman Nayak

Home About Academics Skill Projects



Hi, I am,

Suman Nayak

Master of Computer Application

suman.work.in@gmail.com

+91 7077262619

[linkedin.com/in/suman-nayak-b7b582391/](https://www.linkedin.com/in/suman-nayak-b7b582391/)

github.com/suman-nayak

About Me

Hello! I am Suman Nayak, an enthusiastic and motivated technology learner with a strong interest in Cloud Computing, DevOps, Python, and Web Development. I enjoy learning how modern applications are developed, deployed, and managed using cloud technologies and automation tools. I have been building my skills in AWS, Linux, Python, HTML, CSS, Git, CI/CD, and Infrastructure as Code, while working on practical projects to strengthen my technical knowledge. I am a quick learner, problem solver, and continuous learner who enjoys taking on new challenges and improving my skills through hands-on practice. My goal is to begin my career in a Cloud/DevOps or Software Development role, contribute to real-world projects, and grow into a skilled technology professional. Outside of coding, I believe in continuous improvement, teamwork, and staying curious about emerging technologies.

[Download CV](#)



My Academic Records

MCA
Silicon University
6.91 (CGPA)
2025 - 2027

B.Sc(Physics)
F.M University
7.71 (CGPA)
2021 - 2024

Higher Secondary
Bhadrak Autonomous College
77.05%
2019 - 2021

Secondary
A.C.B.N Bhadrak
79.09%
2019

My Technical Skills

Python	80%	HTML	90%
C	75%	CSS	65%
Java	72%	JavaScript	55%
PHP	78%	Bootstrap	85%

My Projects

Real-Time Weather Dashboard
Tools: Java 21, Java Swings, Maven, Java HttpClient, OpenWeatherMap API, PostgreSQL
A dynamic desktop application that integrates with RESTful APIs to fetch and display live meteorological data. Engineered using JavaSwing for a responsive, modern UI and Java's native HttpClient for efficient network requests. Implemented JSON parsing to extract complex data structures, mapping live metrics (temperature, humidity, wind speed) to visual UI components and condition-based dynamic backgrounds.

Secure Two-Tier Application Infrastructure on AWS
Tools: AWS, Amazon EC2, Amazon VPC, AWS Client VPN Endpoint, ACM, OpenVPN, PostgreSQL
Your organization wants to host a web application on AWS. The web server should be accessible from the internet, while the database server must remain private and inaccessible from the public internet. Administrators should be able to securely connect to the private database from their local machines using a VPN.

E-Commerce Website
Tools: Python, Django, SQLite, HTML & CSS, Bootstrap, JavaScript
This project is a comprehensive, full-stack e-commerce web application designed to simulate a real-world online shopping experience. It features a fully responsive, custom-themed user interface and handles the complete customer journey. Core functionalities include secure user authentication, dynamic product browsing, real-time cart management, and order history tracking. Additionally, it integrates a mock UPI payment gateway to seamlessly demonstrate the checkout and order-completion lifecycle.

Data Redundancy Removal System Design
Tools: Python, Flask, SQLite, Pandas, Git & GitHub
It acts as a smart gatekeeper for your database. When data arrives, the system extracts specific identifying fields and generates a unique cryptographic hash. By enforcing a strict database constraint on this hash, the pipeline accurately classifies incoming data as either unique, a "false positive" (similar data from a different source), or a true redundant duplicate. This ensures your database remains 100% accurate, highly efficient, and completely free of duplicate bloat.

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Mobile view: